

The Density

Zero-Density Estimates and the BN Residual

Diego Rincón

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Abstract

The permanent residual $d_\infty^2 = \lim_{N \rightarrow \infty} d_N^2$ is zero under RH and positive under $\neg$ RH. We quantify its size: $d_\infty^2 = \sum_{\rho: \text{Re}(\rho) > 1/2} |\langle 1_{(0,1)}, \phi_\rho \rangle|^2$ is a sum over off-critical zeros, convergent under any standard zero-density hypothesis. The density function $N(\sigma, T)$ controls the size and structure of this sum. Under the classical density estimate, d_∞^2 may be large; under the density hypothesis, it is small; under RH, it is exactly zero. The dichotomy: $d_N^2 \rightarrow 0$ if and only if the density is zero for all $\sigma > 1/2$ — which is RH.

1 The Permanent Residual

Definition 1 (Zero-density function). For $\sigma \in (1/2, 1)$ and $T > 0$:

$$N(\sigma, T) = \#\{\rho : \zeta(\rho) = 0, \text{Re}(\rho) \geq \sigma, 0 < \text{Im}(\rho) \leq T\}.$$

Under RH: $N(\sigma, T) = 0$ for all $\sigma > 1/2$. Classical zero-density theorem: $N(\sigma, T) = O(T^{A(\sigma)(1-\sigma)+\varepsilon})$ with $A(\sigma) \leq 4$ (Ingham, 1940). Density hypothesis: $A(\sigma) \leq 2$ for all σ .

Definition 2 (Off-critical orthogonal functions). For each zero $\rho_0 = \sigma_0 + i\gamma_0$ of ζ with $\sigma_0 > 1/2$, the function ϕ_{ρ_0} defined by its Mellin transform $\hat{\phi}_{\rho_0}(s) = 1/(s - \rho_0)$ (up to normalization) is in $\mathcal{V}_\infty^\perp$, the orthogonal complement of the BN span in $L^2(0, 1)$ ([1]). Its inner product with $1_{(0,1)}$ is:

$$\langle 1_{(0,1)}, \phi_{\rho_0} \rangle = \frac{1}{2\pi i} \int_{\text{Re}(s)=1/2} \frac{1}{s} \cdot \frac{1}{s - \rho_0} |ds| = \frac{1}{\rho_0(\rho_0 - 1)}.$$

Theorem 3 (The permanent residual formula). *Under $\neg$ RH:*

$$d_\infty^2 = \lim_{N \rightarrow \infty} d_N^2 = \sum_{\substack{\rho: \zeta(\rho)=0 \\ \text{Re}(\rho) > 1/2}} \frac{c_\rho}{|\rho|^2 |\rho - 1|^2}$$

where $c_\rho > 0$ are computable constants depending on the multiplicity and position of ρ . In particular, $d_\infty^2 > 0$ whenever there is at least one off-critical zero.

Proof sketch. The BN flow f_N^* converges in $L^2(0, 1)$ as $N \rightarrow \infty$ to $f_\infty^* = \pi_{\mathcal{V}_\infty}(1_{(0,1)})$, the projection onto the BN span. The residual $r_\infty = 1_{(0,1)} - f_\infty^*$ is orthogonal to $\mathcal{V}_\infty$. By [1], $\mathcal{V}_\infty^\perp$ is spanned by the functions ϕ_ρ for off-critical zeros ρ . Expanding r_∞ in the ϕ_ρ basis and using Parseval gives the formula for $d_\infty^2 = \|r_\infty\|^2$. $\square$

Corollary 4 (Dichotomy).

$$d_\infty^2 = 0 \iff RH.$$

2 Density Controls the Residual

Theorem 5 (Density bounds the residual). *Assume $N(\sigma, T) = O(T^{A(1-\sigma)+\varepsilon})$ for some $A \geq 2$. Then:*

$$d_\infty^2 \leq C_A \cdot \sum_{\rho: \operatorname{Re}(\rho) > 1/2} \frac{1}{|\rho|^2} \leq C'_A \int_{1/2}^1 \int_1^\infty t^{-2} dN(\sigma, t) d\sigma.$$

This integral converges: by integration by parts in t ,

$$\int_1^\infty t^{-2} dN(\sigma, t) = [t^{-2} N(\sigma, t)]_1^\infty + 2 \int_1^\infty t^{-3} N(\sigma, t) dt = O\left(\int_1^\infty t^{A(1-\sigma)-3+\varepsilon} dt\right),$$

which converges for $A(1-\sigma) < 2$, i.e., $\sigma > 1 - 2/A$. For $A = 4$ (classical): converges for $\sigma > 1/2$. For $A = 2$ (density hypothesis): converges for $\sigma > 0$.

Corollary 6 (Density hypothesis bound). *Under the density hypothesis ($A = 2$ above):*

$$d_\infty^2 \leq C \int_{1/2}^1 \int_1^\infty t^{2(1-\sigma)-3+\varepsilon} dt d\sigma = C \int_{1/2}^1 \frac{d\sigma}{2\sigma - \varepsilon} = O(1).$$

The residual is finite and bounded, but may be > 0 . Under RH, all terms in the sum vanish and $d_\infty^2 = 0$.

Remark 7 (What density cannot do). The density hypothesis controls how MANY zeros can lie off the line, but it cannot prevent even one zero from being off the line. Since a single zero ρ_0 contributes $c/|\rho_0|^2 > 0$ to d_∞^2 , the density hypothesis alone cannot prove $d_\infty^2 = 0$. To prove $d_\infty^2 = 0$, we need $N(\sigma, T) = 0$ for all $\sigma > 1/2$ — RH. This is the precise sense in which density is not enough.

3 The Rate of Approach to d_∞^2

Under $\neg RH$, the flow converges to $d_\infty^2 > 0$. The rate of this convergence is controlled by the real parts of the off-critical zeros.

Proposition 8 (Rate of convergence to permanent residual). *Let $\sigma_1 = \inf\{\operatorname{Re}(\rho) : \zeta(\rho) = 0, \operatorname{Re}(\rho) > 1/2\}$ be the real part of the off-critical zero closest to the line. Then:*

$$d_N^2 - d_\infty^2 = O(N^{-(2\sigma_1-1)+\varepsilon}).$$

Under RH, $\sigma_1 = 1/2$ (or $+\infty$ if no off-critical zeros exist), and the formula reduces to $d_N^2 = O(N^{-c/\log \log N})$ as in [3].

Proof sketch. The transient part $d_N^2 - d_\infty^2$ measures how much the flow has still not converged to its limit. By the Verblunsky analysis of [2], the slowest-decaying term in $d_N^2 - d_\infty^2$ comes from the zero ρ_1 with $\text{Re}(\rho_1) = \sigma_1$: the correction to the Verblunsky coefficients at step N decays as $N^{-(\sigma_1 - 1/2) + \varepsilon} = N^{-(2\sigma_1 - 1)/2 + \varepsilon}$, and squaring gives the stated rate. $\square$

Remark 9 (The ladder of rates). The rate $N^{-(2\sigma_1 - 1)}$ forms a ladder:

$$\begin{aligned} \sigma_1 = 1/2 + 1/2 = 1 : \quad d_N^2 - d_\infty^2 &= O(N^{-1 + \varepsilon}) \\ \sigma_1 = 3/4 : \quad d_N^2 - d_\infty^2 &= O(N^{-1/2 + \varepsilon}) \\ \sigma_1 = 1/2 + \varepsilon : \quad d_N^2 - d_\infty^2 &= O(N^{-\varepsilon}) \end{aligned}$$

As $\sigma_1 \rightarrow 1/2^+$, the convergence slows. Under RH ($\sigma_1 > 1/2$ for no zero), $d_\infty^2 = 0$ and the rate from [3] takes over.

4 The Density Spectrum

Definition 10 (Density spectrum of the BN flow). The density spectrum is the function:

$$\Lambda(\sigma) = \sum_{\rho: \text{Re}(\rho) \geq \sigma} \frac{c_\rho}{|\rho|^2 |\rho - 1|^2}, \quad \sigma \in (1/2, 1),$$

which records how much of d_∞^2 comes from zeros with real part $\geq \sigma$. Λ is decreasing, $\Lambda(1) = 0$, $\Lambda(1/2^+) = d_\infty^2$.

Theorem 11 (Density spectrum and zero-free regions). *The following are equivalent:*

1. $\Lambda(\sigma_0) = 0$ for some $\sigma_0 > 1/2$: there are no zeros with $\text{Re}(\rho) \geq \sigma_0$.
2. $d_N^2 = O(N^{-(2\sigma_0 - 1) + \varepsilon})$: the flow converges to 0 at polynomial rate.
3. The zero-free region includes $\{\text{Re}(s) \geq \sigma_0\}$.

In particular, $\Lambda(1/2^+) = 0 \iff \text{RH}$.

Remark 12 (The density spectrum is the wall). The density spectrum Λ is the exact measure of the BN wall. Its value at $\sigma = 1/2$ is the permanent residual. To prove $\Lambda(1/2^+) = 0$, one must prove no zeros exist off the line — which is precisely RH. Every existing partial result (zero-free regions $\sigma > 1 - c/\log T$, density estimates $N(\sigma, T) = O(T^A)$) only says Λ is small for σ bounded away from $1/2$. The limit $\sigma \rightarrow 1/2$ is the wall, and no classical tool reaches it.

Theorem 13 (The BN flow as zero detector). *The BN flow is an optimal detector for off-critical zeros in the following sense:*

$$d_\infty^2 = 0 \iff \text{no off-critical zeros exist,}$$

and the value $d_\infty^2 > 0$ encodes exactly which zeros are off the line and where. Moreover, the permanent residual r_∞ is a superposition of the functions ϕ_ρ for each off-critical zero: $r_\infty = \sum_\rho \alpha_\rho \phi_\rho$. Knowing r_∞ is equivalent to knowing all off-critical zeros and their multiplicities.

Proof. The Parseval identity $d_\infty^2 = \|r_\infty\|^2 = \sum_\rho |\alpha_\rho|^2 \|\phi_\rho\|^2$ gives the equivalence. Each $\alpha_\rho = \langle r_\infty, \phi_\rho \rangle / \|\phi_\rho\|^2$ encodes the contribution of ρ to the permanent residual. Recovering all α_ρ from r_∞ (via inner products) gives all off-critical zeros. $\square$

Remark 14 (The permanent residual as a witness). Theorem 13 says: r_∞ is a witness for $\neg\text{RH}$. If RH fails with zero ρ_0 , then $r_\infty \neq 0$ and $\langle r_\infty, \phi_{\rho_0} \rangle \neq 0$: the residual “knows” about ρ_0 . The difficulty is not detecting the witness; it is showing no witness exists. To show $r_\infty = 0$, one must show $d_N^2 \rightarrow 0$, which is RH. The witness can be found if $\neg\text{RH}$, but proving it doesn’t exist requires proving RH directly.

References

- [1] Rincón, D. (2026). Möbius. Preprint.
- [2] Rincón, D. (2026). Verblunsky. Preprint.
- [3] Rincón, D. (2026). The Rate. Preprint.
- [4] Rincón, D. (2026). Wisdom. Preprint.
- [5] Ingham, A. E. (1940). On the estimation of $N(\sigma, T)$. *Quarterly Journal of Mathematics*, 11, 291–292.
- [6] Báez-Duarte, L. (2005). A strengthening of the Nyman–Beurling criterion. *Atti della Accademia Nazionale dei Lincei* 14(1), 5–11.